

AI/ML Internship – Day 65 Notes & Tasks

Module 8: Building an AI Chatbot with Conversation Memory

Part 1: Recap of Day 64

Yesterday we learned:

-  What is an API?
-  OpenAI API
-  API Key
-  AI Chatbot using Python
-  Sending user messages to an AI model

Today we will learn:

- What is Conversation Memory?
 - Why memory is important
 - How AI chatbots remember conversations
 - Building a chatbot with chat history
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Part 2: What is Conversation Memory?

Definition

Conversation Memory is the ability of a chatbot to remember previous messages and use them while answering future questions.

Simple Definition

Conversation Memory allows the chatbot to continue a conversation naturally instead of treating every question as a completely new conversation.

Part 3: Why is Conversation Memory Important?

Imagine this conversation:

You: My name is Rahul.

Bot: Nice to meet you Rahul!

You: What is my name?

Without memory:

Bot:

Sorry, I don't know your name.

With memory:

Bot:

Your name is Rahul.

This is called **Conversation Memory**.

Part 4: Real-Life Example

Customer:

I want to book a flight.

Bot:

Which city do you want to travel to?

Customer:

Delhi

Bot:

When do you want to travel?

Customer:

Tomorrow

Bot:

Your flight to Delhi for tomorrow is being searched.

Notice that the chatbot remembers:

- Flight
- Delhi

- Tomorrow

📌 Part 5: Chatbot Workflow with Memory

User Message

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Store Previous Messages

|



Send Chat History

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AI Model

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Generate Context-Aware Response

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Display Response

📌 Part 6: Chat History

Instead of sending only the latest message, send the **entire conversation**.

Example:

User:

Hello

Assistant:

Hi! How can I help you?

User:

My name is Rahul.

Assistant:

Nice to meet you Rahul.

User:

What is my name?

The AI can now answer correctly because it has the previous conversation.

Part 7: Python Example (Concept)

```
conversation = []
```

```
conversation.append({  
    "role": "user",  
    "content": "Hello"  
})
```

```
conversation.append({  
    "role": "assistant",  
    "content": "Hi! How can I help you?"  
})
```

```
conversation.append({  
    "role": "user",  
    "content": "My name is Rahul"  
})
```

The list stores the complete conversation.

🚀 Part 8: AI Chatbot with Memory

```
from openai import OpenAI
```

```
client = OpenAI(api_key="YOUR_API_KEY")
```

```
conversation = []
```

```
print("🤖 AI Chatbot")
```

```
print("Type 'bye' to exit.")
```

```
while True:
```

```
    user = input("You: ")
```

```
    if user.lower() == "bye":
```

```
        print("Bot: Goodbye!")
```

```
        break
```

```
    conversation.append({
```

```
        "role": "user",
```

```
        "content": user
```

```
    })
```

```
    response = client.responses.create(
```

```
        model="gpt-4.1-mini",
```

```
        input=conversation
```

```
)
```

```
answer = response.output_text
```

```
print("Bot:", answer)
```

```
conversation.append({  
    "role": "assistant",  
    "content": answer  
})
```

Part 9: Example Conversation

You:

My favourite language is Python.

Bot:

Python is a great programming language.

You:

Which language do I like?

Bot:

You like Python.

Without memory:

Bot:

I'm not sure.

Part 10: Memory vs No Memory

Without Memory

Every question is independent

No context

Less natural

Limited interaction

With Memory

Previous messages are remembered

Understands conversation context

More human-like

Continuous conversation

Part 11: Applications

Customer Support

Remember customer issues during the conversation.

Healthcare

Remember patient symptoms while chatting.

Banking

Remember account-related queries in the current session.

Education

Remember the student's previous questions.

E-commerce

Remember shopping preferences and previous requests.

Part 12: Advantages

 Natural conversations

 Better user experience

- ✓ Context-aware responses
 - ✓ Personalized interaction
 - ✓ Reduces repeated questions
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✚ Part 13: Limitations

- ✗ Uses more tokens (higher API usage)
 - ✗ Long conversations may become expensive
 - ✗ Older messages may need summarization
 - ✗ Developers should avoid storing sensitive information unnecessarily
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✚ Part 14: Best Practices

- Keep only relevant conversation history.
 - Remove unnecessary old messages.
 - Never store passwords or confidential information.
 - Summarize long conversations if needed.
 - Use memory only when it improves the user's experience.
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✚ Part 15: Theory Questions

What is Conversation Memory?

Conversation Memory is the chatbot's ability to remember previous messages and use them in future responses.

Why is conversation memory important?

It enables natural, context-aware conversations and improves the user experience.

What is chat history?

Chat history is the collection of previous messages exchanged between the user and the chatbot.

Why should developers manage conversation history?

To reduce API costs, improve performance, and protect user privacy.

Give three applications of conversation memory.

- Customer Support
 - Education
 - Healthcare
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Part 16: Interview Questions

What is conversation memory in AI chatbots?

It is the ability to retain previous messages and use them to generate context-aware responses.

How does a chatbot remember previous messages?

By storing the conversation history and sending it along with new user messages.

What are the benefits of memory in chatbots?

Improved context, personalized responses, and more natural conversations.

What is a challenge of maintaining conversation history?

Long conversations consume more tokens and increase API usage.

How can developers optimize long conversations?

By summarizing older messages and keeping only important context.

Tasks for Day 65

Theory

1. Define Conversation Memory.
 2. Explain why memory is important in AI chatbots.
 3. Explain the workflow of a chatbot with memory.
 4. List five advantages of conversation memory.
 5. Differentiate between chatbots with and without memory.
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Practical Task 1

Create a Python list named conversation and store five messages in it.

Practical Task 2

Modify your Day 64 chatbot to save user messages and assistant responses.

Practical Task 3

Test your chatbot with this conversation:

My name is Rahul.

I am from Kerala.

What is my name?

Where am I from?

Observe whether it answers correctly.

Practical Task 4

Draw the workflow of a chatbot with conversation memory.

Practical Task 5

List five real-world applications where conversation memory is useful.

